

Communication between MES and PLC

1. General

Naming Conventions

The following naming conventions are utilized throughout the integration documentation:

- **Mill System (MES):** The central manufacturing execution server.
- **PLC:** The programmable logic controller.
- **Label Printer (LBL):** The physical printing hardware for product identification.

PLCs, PCs, Names, IP-addresses and Ports

Name	IP-address	Port(s)	Notes
MES			
PLC			
Labeling printer	Installed as a Windows printer	—	

- **Mask:** 255.255.255.0
- **Gateway:** Place

2. Interface and Communication Details

- **Network Type:** Ethernet
- **Protocol:** TCP/IP Sockets
- **Data Transfer Type:** TCP Data Streams (ASCII-strings)
- **Message Type:** ASCII-string with separator character between data fields.

General Specifications

- All data fields must be readable and printable ASCII-characters.
- Data fields are separated by a pipe ("|") character (ASCII 124 / HEX 7C).
- Length of the message (240 chars) must be the same in each message (MES ↔ PLC).
- Numeric (integer, float) values must be converted to ASCII-characters.
- Decimal numbers (digits after decimal point) must be separated by a dot (".").
- Character data fields are padded with trailing blanks.
- Numeric data fields are padded with leading zeros.
- Message starts with a Start of TeXt <STX> (HEX 02).
- Message ends with an End of TeXt <ETX> (HEX 03).
- **Message Buffering:** No message buffering, i.e. link down time data will not be sent later.
- **Data Format:** <STX>MessageID|Message|Data1|Data2|...|Data n|<ETX>

- Message ID (4 chars) = Message ID of Message
- Example: "WR01|0010|01|0000987777|...|"

Connection Architecture

- **Client (active):** PLC
- **Server (listen):** MES
- Client side opens the connection and maintains it with **KEEP_ALIVE** messages, so there is no need to reopen connection before each transmit.
- If there is nothing to transmit, a **KEEP_ALIVE** message is sent at the specified interval to maintain the link.
- The client side closes and reopens the connection if it fails to transmit messages successfully.

3. Messages when Product at Conveyor Starting Position #1

Operation & Sequence

1. PLC sends a set data query to the MES.
2. MES responds with the set data message.

Communication Flow:

- **PLC -> MES:** (1) Sends RW01 (Set Data Query)
- **MES -> PLC:** (2) Sends MW01 (Set Data at stopper)

3.1 Communication

3.2 "RW01" Set Data Query (MES ↔ PLC)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	RW01	
—	Delimiter	1		HEX 7C (Pipe)
—	End of MSG Char	1	ETX	HEX 03

3.3 "MW01" Set Data at stopper (MES ↔ PLC)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	MW01	
—	Delimiter	1		HEX 7C (Pipe)
2	Product ID (Unit ID)	12	0123456789aa	
—	Delimiter	1		HEX 7C (Pipe)
—	End of MSG Char	1	ETX	HEX 03

No	Description	Length	Format	Comment

4. Messages when Product at Scale Conveyor

Operation

1. PLC measures product weight.
2. PLC sends the product id and measured weight to MES.
3. MES checks that measures are within tolerances.
4. MES sends an acknowledgement message.

Communication Flow:

- **PLC -> MES:** (1) "RG01" Roll at scale conveyor (Product ID & Measured Weight)
- **MES -> PLC:** (2) "MG01" Roll data acknowledgement

"RG01" Roll is identified at scale conveyor (MES ↔ PLC)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	RG01	
—	Delimiter	1		HEX 7C (Pipe)
2	Product ID (Unit ID)	12	0123456789af	
—	Delimiter	1		HEX 7C (Pipe)
3	Product weight	4	0000-9999	[kg]
—	Delimiter	1		HEX 7C (Pipe)
—	End of MSG Char	1	ETX	HEX 03

"MG01" Roll Data acknowledgement (MES ↔ PLC)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	MG01	
—	Delimiter	1		HEX 7C
2	Roll ID (Unit_ID)	12	0123456789aa	
—	Delimiter	1		HEX 7C
—	End of MSG Char	1	ETX	HEX 03

5. Messages when Product at Labeling Station

Operation

1. PLC sends the product id and printer number to MES.
2. MES prints the label via the local Windows driver.
3. MES sends a message to PLC informing that the label is printed or reporting an error.
4. PLC sends a request for each required label.

Communication Flow:

- **PLC -> MES:** (1) "RE01" Roll at Labelling station
- **MES -> LBL:** (2) MES prints label if roll found
- **MES -> PLC:** (3) "ME01" Roll Label Printed

"RE01" Roll is identified at Labeling station (MES ↔ PLC)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	RE01	
—	Delimiter	1		HEX 7C (Pipe)
2	Roll ID (Unit_ID)	12	0123456789af	
—	Delimiter	1		HEX 7C (Pipe)
3	Label printer	1	0-1	Printer number
—	Delimiter	1		HEX 7C (Pipe)
—	End of MSG Char	1	ETX	HEX 03

"ME01" Roll label printed (MES ↔ PLC)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	ME01	
—	Delimiter	1		HEX 7C (Pipe)
2	Roll ID (Unit_ID)	12	0123456789aa	
—	Delimiter	1		HEX 7C
3	Status	1	0-1	OK, FAULT
—	Delimiter	1		HEX 7C (Pipe)
—	End of MSG Char	1	ETX	HEX 03

6. Other Messages

Keep Alive Message

- PLC sends a keep alive message every 120 seconds.
- If a reply message is not received from MES within 20 seconds, the connection is restarted.
- The message counter is incremented after every message.

"RR01" Keep Alive Message (PLC → MES)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	RR01	
—	Delimiter	1		HEX 7C
2	Message counter	4	123	0001 – 9999
—	Delimiter	1		HEX 7C
—	End of MSG Char	1	ETX	HEX 03

Keep Alive Acknowledge Message

- MES must reply with the following message within 20 seconds of receipt.
- The client follows the message counter set by the RRHS process.

"MS05" Keep Alive Acknowledge Message (MES → PLC)

No	Description	Length	Format	Comment
—	Start of MSG Char	1	STX	HEX 02
1	Message ID	4	MS05	
—	Delimiter	1		HEX 7C
2	Message counter	4	123	0001 – 9999
—	Delimiter	1		HEX 7C
3	Spare	227	'All spaces'	HEX 20 (437 spaces)
—	Delimiter	1		HEX 7C
—	End of MSG Char	1	ETX	HEX 03